ABSTRACT

This study examines the challenges faced by communication signals as they travel through outer space, focusing on the effects of gravity in high-mass-density environments such as Jupiter. It investigates the main phenomena that hinder signal transmission, most notably gravitational shift, which leads to signal energy loss, and time delay results from signal path curvature and time flow variation, in addition to signal distortion due to gravitational lenses. This research also aims to review engineering solutions to overcome these challenges, which include designing adaptive communication systems capable of compensating for energy loss, using advanced error correction techniques and developing intelligent algorithms to address communication problems. The research concludes by presenting the practical application that have been implemented, such as the Global positioning system (GPS) as the most prominent model that has been applied to overcome these problems highlighting the role of communications engineers in ensuring the continuity and effectiveness of satellite communications under the influence of these harsh conditions